



I would walk 500 miles (if it paid): Vouchers and school choice in Chile[☆]

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ABSTRACT

One of the pillars of the educational voucher system is that competition among schools to attract students would improve the quality of the education. Surveys to parents and previous work suggested that families rank the distance of the school from their home as the most important factor for choosing a school. They also suggest that parents largely ignore the results of standardized tests. We use a novel data-set which includes precise measures of the distance between homes and schools to analyze the determinants of school choice in Chile. Economic theory suggests, and the estimations confirm, that parents consider quality and the location when choosing schools. The paper quantifies the relevant trade-offs.

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1. Introduction

A number of countries have implemented a voucher system, but in none is as widespread as in Chile. While most other countries conducted small experiments in some cities or groups of schools, the Chilean case stands out because it established a country-wide voucher system. Moreover, in opposition to other experiences, Chile's

voucher system is very much in line with Friedman's (1955) original idea because financing does not distinguish between public and private providers. In 2010, over 90% of the Chilean students attending public and private schools received funding via vouchers (Paredes & Ugarte, 2011).

However, the debate regarding the convenience of adopting a voucher system is intense.¹ Ladd (2002) makes an extensive review of different experiences, both in the US and abroad, and concludes that at best, the gains of a large and massive voucher program are small. In a report on the performance of the Washington-DC Opportunity

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¹ When John McCain supported the voucher system as the Republican candidate for the US presidency, professor Carnoy sent a public letter to Senator McCain stating: "... So I don't get why you are advocating vouchers as a core educational policy you would push in your Administration. If they have no positive effect on low-income pupils' achievement, why bother? Sure, choice is nice, but if there is no evidence students learn more, why keep insisting that choice 'works'?" ("Your Education Policies Don't Figure: An Open Letter to John McCain").

Scholarship Program (OSP) for low-income voucher students, Wolf et al. (2008) do not find that private voucher schools perform better than similar students in public schools.

A key issue to understand the potential effect of the voucher system is related with the conditions in which it operates. The Chilean voucher system was designed assuming that the parents' right to choose their children's school would cause competition among schools and improve the quality of the education provided. Parents would reject low quality schools, which would have a declining number of students, and eventually would be eliminated from the market.

This crucial part of the theory underlying the voucher system has been questioned by several studies. For example, Carnoy and McEwan (2000), McEwan and Carnoy (2000), and Elacqua and Fabrega (2004) consider that the relatively poor results attributed to the Chilean voucher system are due to the parent's lack of capacity for choosing a school. This view is supported by surveys (e.g., CEP, 2006), that show that less than 50% of parents knew the results of the standardized tests of their children's school and less than 1% considered these results as the most important factor in choosing their children's school. The most frequent response about the most important factor behind the choice of a school was its proximity to the home (25.2%). If correct, this evidence would undermine a crucial feature that is necessary for fostering quality through the voucher system.

This paper constructs and uses a data-set that includes precise measurements of distance and it develops a simple specification for addressing the choice of school incorporating distance, price and quality. It evaluates how important distance is (as compared to other factors) by focusing on how parents actually choose schools (revealed preferences) and not on results of surveys that merely enunciate factors that they consider important.

The paper is organized as follows. Section 2 describes the Chilean voucher system and the main findings regarding it in the literature. Section 3 describes the data-set and presents some descriptive statistics. Section 4 discusses the econometric framework used to evaluate the determinants of the choice of school and reports the results. Finally, Section 5 concludes.

2. The Chilean voucher system

Chile introduced a number of pioneering reforms in the last quarter of the past century (Chumacero, Fuentes, Lüders, & Vial, 2007). A first wave of basic structural reforms included price liberalization, the reduction of import tariffs, and an overall reduction of the role of the government in the economy. A second wave of reforms "revolutionized" the social security regime shifting from a pay-as-you-go to a privately managed fully funded system, privatized most of the State-owned-enterprises (including public utilities), and introduced the voucher system in education. The Chilean experience constituted a referent for reforms conducted later elsewhere (Aedo & Sapelli, 2001).

Chile's per capita GDP growth was way below the average of East Asia, OECD countries, and the world economy prior to the reforms. After them, and particularly in the past 25 years, Chile's economic performance has been outstanding and is the only country in Latin America that can claim significant progress in reducing income gaps with the developed world and reducing the level of poverty in more than 50%.

Despite these achievements, there are two areas in which little progress has been made. The first is income distribution: Chile presents one of the most unequal distributions in the region and it has remained virtually unchanged. The second is in improving education quality: despite huge increases in the budget allocated to this sector, Chile's performance is relatively poor when compared with similar countries.

The last issue is particularly relevant, given that, since 1981 a nation-wide voucher system has been in place. Public schools were transferred to municipalities, decentralizing primary and secondary education. Since then, school administration has been carried out by both public and private providers. As a result of this policy, there are basically three types of schools; municipal, private subsidized (subsidized) and entirely private. The first two receive most of their funds from State subsidies and, in the case of the privately subsidized schools, may also receive contributions by the parents. The entirely private schools do not receive State subsidies. The State funding is conducted through a scheme of educational grants or subsidies per student (demand subsidies), which are to contribute to operating expenses and capital facilities.

There are three important differences between schools. The first is that the administration of subsidized schools is mainly done by the private sector, while the administration of municipal schools, as its name implies, is the responsibility of municipalities. Second, municipal schools are required to accept any student, while private schools may select children based on their academic achievements.² The third difference is that the payment of municipal schools depends entirely on the State subsidy, while some subsidized schools require a surcharge to parents. The amount of the subsidy depends on the hours of classes taught and teaching method (e.g. nursery, primary or secondary levels) and, by 2010 was of about US\$ 94 per month. In the case of private schools they are funded entirely through a monthly payment by parents and guardians, in a relationship that is governed by a contract for the provision of educational services.

According to statistics released by the Ministry of Education in 2010, there were 5847 municipal schools, 5263 subsidized schools, and 727 private schools. In the Metropolitan Region, where Santiago is located, there were 750 municipal schools, 1755 subsidized schools, and 340 private schools. In terms of enrollment, the share of students attending paid private schools has remained virtually unchanged since 1981, but the enrollment in subsidized schools administered privately has increased

² However, successful municipal schools at the secondary level select students based on academic achievements.

steadily; thus, decreasing the share of enrollment in public schools (Paredes & Pinto, 2009).

The system has achieved almost universal coverage, and impressive results in terms of increasing the number of schools and their infrastructure. However, and despite the three fold real increase in public expenditure in education over the last 20 years, by international comparisons, the results in terms of quality are rather disappointing.

As documented in several studies, Chile performed poorly in international tests such as TIMSS and PISA before 2009 (Bellei & González, 2004; Brunner, 2005; Heyneman, 1991, 2004). For instance, the TIMSS-R (Third International Study of Sciences and Mathematics) shows that Chilean eighth graders ranked 35th among 38 countries both in mathematics and science in 1999. In 2003, they ranked 40th and 37th among 45 countries in math and science tests respectively. Chile underperforms even when compared to countries with lower per capita GDP and lower investment in education, such as Jordan and Malaysia. Furthermore, the TIMSS-R 2003 shows that the gap in the level of quality between the low and high income sectors increased between 1990 and 2003 from 120 to 142 points on a scale with a mean of 500 and a standard deviation of 100.³ In addition, the main tool used to measure and compare schools' results in Chile, a standardized test named SIMCE that is applied at a national level and a yearly basis,⁴ shows the existence of a large difference between schools with students coming from low income families and schools with wealthier students.

The apparent paradox regarding the persistently poor quality of education in a context of otherwise successful reforms has generated debate and an extensive literature with respect to the Chilean experience. Drago and Paredes (2011) review the results of papers that use alternative methods to estimate performance functions for Chile. In the line with the international literature, all the studies find that conventional socio economic variables (such as income, parent schooling, and gender) are relevant.⁵ After controlling for other factors, on average, municipal schools have lower scores than the other two types of establishments.

There is also a strand of the literature that has focused on the issue of selection and its effects. Elacqua, Schneider, & Buckley (2006) analyze school selection made by parents, finding that academic results are crucial in parents' school choice. Gallego and Hernando (2008) conclude that selection is basically made by parents and not by schools, and that the selection made by parents is restricted because of the importance of the location of the schools. Contreras, Elacqua, & Salazar (2007) use dummy variables for those

schools that select students to conclude that those schools tend to have better results. García and Paredes (2010) use a different approach to control for selection, correcting for sample selection. They estimate that the effect of selection on academic performance is statistically significant, but at most, 1/10 of a standard deviation of the SIMCE test. They attribute this finding to the high competition to keep students among schools, particularly in a context of free entry.

Closely linked with the idea that incentives also matter in the education area, an important line of the literature analyzes the effect of competition on results. For the USA, Hoxby (1994) combined five data-sets which are matched geographically and centered on the year 1980. She finds that greater private school competition significantly raises the quality of public schools, as measured by the educational achievement, wages, and high school graduation rates of public school students. In addition, she finds some evidence that public schools react to greater competition of private schools by paying higher salaries to teachers. Hoxby (2000) finds evidence that easier choice leads to greater productivity. Areas with greater opportunities for choice among public schools have lower per-pupil spending, lower teacher salaries, and larger classes. The same areas have better average student performance, as measured by students' educational achievement, teachers' wages, and test scores.

Rouse (1998), referring only to Milwaukee, is less emphatic suggesting that the Parental Choice Program seems to have had a positive effect on math achievement in attending a private school, but had no benefits for reading scores. Bayer and McMillan (2005) measure competition as an elasticity which represents how a reduction in quality would affect school demand. They find a significantly positive relationship between competition and scores with data of an urban area in the USA (see, Braun-Munzinger, 2005, for a review). On the theoretical side, Epple and Romano (1998, 2002) develop a model of competition between private and public schools in a voucher system. They conclude that this system promotes the growth of the private sector, increases sorting, and tends to benefit high-ability students.

The effects of competition on incentives and performance have also been studied for the Chilean case. Hsieh and Urquiola (2003) construct a panel with data from 150 counties between 1982 and 1996, and measure the difference in schools' results before and after the 1981 reform. They compare the changes among rural and urban schools assuming that competition, measured by the entrance of the new voucher-funded private schools, was less intense in rural than in urban areas. They conclude that the voucher system did not improve school performance and that, in turn, produced sorting. They also argue that competition may have not had an effect on performance because of the way in which parents choose schools. Gallego (2006) tests the existence of a positive relationship between competition and school performance, and whether this relationship is more important for private voucher schools than for public schools. Using cross-section regressions to explain SIMCE scores with a county-level competition index measured as the proportion of students in each county that

³ Based on other international tests such as PISA, IALS, Kimko and TIMSS, Tokman (2004) arrive at similar conclusions.

⁴ Until 2005, the application of the tests was alternated between 4th Grade Elementary, 8th Grade Elementary and 2nd Year Junior High. As of the year 2006, 4th Grade Elementary is evaluated every year while alternating between 8th Grade Elementary and 2nd Year Junior High.

⁵ These results are in line with international studies such as Hanushek (1997), Currie and Duncan (1995), Corman (2003), Baker, Brian, and Letendre (2003) and Kan and Tsai (2005).

attend private schools, and other socioeconomic variables, he finds support for both hypotheses. [Auguste and Valenzuela \(2004\)](#) also find a positive effect of competition on schools' results and conclude that competition increases the sorting of students between public and private schools based on students' family income.

More directly related with parents choice and distance, using "a natural experiment and a field experiment" in the context of the No-Child-Left-Behind Act for Charlotte-Mecklenburg Public School District, [Hastings and Weinstein \(2007\)](#) conclude that information is a relevant variable when parents choose schools. They observe family choices submitted in the Spring of 2004, when they had no direct information on school test scores, and then, the choices they submitted in July 2004 after receiving the NCLB-mandated test score information. After receiving NCLB information, the fraction of parents that choose an alternative school increased by 5.1 percentage points relative to a base of 11%, and 14.5 percentage points including only parents who chose NCLB schools first. In both cases, the average score test of the selected school increased.

Even though distance is in the center of public policies, only a few studies have directly considered distance to school from the household as a relevant factor in choosing school. [Hastings, Kane, and Staiger \(2006\)](#) estimate a mixed-logit demand model using the rankings of parents' top three school choices in the first year of school choice. They conclude that parents value proximity and schools' average test scores, and that the importance given by parents to scores increases in family income and in student skills. For a student with average preferences, the preference for proximity is very large relative to the preference for school test scores. Thus, an average student with average preferences would choose their home school over a school three miles further away in which average test scores were 1–2 standard deviations higher, *ceteris paribus*. In addition, for the average student receiving lunch subsidies, the difference in school test scores would have to be even larger to induce them to switch. However, they observe a large variance in preferences. While many students are very inelastic with respect to school test scores, there is a significant density of students who highly value school scores and have low preferences for their neighborhood school. These students are willing to consider schools outside of their neighborhood, and they place a high value on average test scores when choosing a school. This work is considered the international base line for our work, since it uses student and school characteristics such as distance from home and average test scores, and also quantifies proximity as an attribute.

Among the studies applied for less developed countries, [Gertler and Glewwe \(1989\)](#) analyze the role of distance for Peru, and [Alderman, Orazem, and Paterno \(2001\)](#) do it for Pakistan. For Chile, [Gallego and Hernando \(2008\)](#) estimate a random utility model and conclude that parents take into account schools' average scores, accessibility (using a rough measure of average distance for districts) and the fees charged by schools. While this is the only direct effort made to analyze parent's choice in Chile through a revealed preference approach, they use a cruder measure of distance, as they define it as the Euclidean distance between

the centroid of the county (*comuna*) of the household and the school that the child attended. When comparing theirs with ours, it can be shown that their measure tends to overestimate the distance in approximately 80% of the cases, and that this discrepancy is significantly related with the income of the household.

Our paper departs from the previous literature in two dimensions: firstly, it uses a novel data-set which measures distance precisely. Secondly, it uses a new way to model school choice that is able to efficiently include distance and performance.

3. The data

Parents consider several factors when choosing a school for their children; distance from the household is one of them. Measuring distance from the household to the schools is not easy as information of both the location of the household and the school that a child attends is not readily accessible.

Administrative records for students that take the standardized tests provide information of the results of their tests, characteristics of the school, and socio-economic characteristics of the household. However, as the address of the household cannot be recovered from these records, measures of distance cannot be obtained.

The route taken in this paper is by using the National Socio-economic Survey (CASEN). This is a cross sectional household survey taken by the Ministry of Planning of Chile (MIDEPLAN) that has detailed information of socio-economic characteristics of Chilean households and on the impact of social policies and programs on life quality. The CASEN survey has statistical representation at the national, regional, urban and rural levels, and at most of the Chilean districts and counties. It is the main tool used for measuring socio-economic variables as is routinely used to design and evaluate social policies. For a statistically representative sample of the total population, CASEN considers expansion weights (regional and municipal), which are calculated from official population projections provided by the National Institute of Statistics (INE). The database used includes the official INE population projections from the 2002 Census and the weights.

The survey used in this study was conducted in the year 2003 on a sample of 73,720 households that is representative at a district level.⁶ As the survey indicates the name of the school attended, we pinned down the precise location of the school through a school directory. To measure the distance from the household to the school one needs to identify the location of the household. Regrettably this cannot be done, as CASEN does not provide a precise address for the household surveyed. However, the survey does contain information about the block in which the household is located.

⁶ The (regional and municipal) expansion factors used in the survey are calculated using the official population projections made by the National Bureau of Statistics (INE). All the estimations conducted in this paper use these expansion factors.

Table 1
Descriptive statistics by type of school administration (averages).

Variable	Total	Municipal	Subsidized	Private
Share of women (%)	49.3	48.8	49.5	49.7
Years of schooling (father)	12.6	10.9	12.5	16.8
Years of schooling (mother)	12.0	10.5	12.1	15.3
Income per capita (US\$ per month)	229.0	123.8	188.3	656.2
Share of students (%)	100.0	33.6	53.1	13.3

Weighted sample: 1,203,799.

We georeferenced these blocks using digital maps provided by the companies DICTUC and Mapcity, which cover the entire city of Santiago, where we focus the analysis. At the same time, using the School Directory 2003 of the Ministry of Education (MINEDUC), which includes all of the schools' addresses, it was possible to georeference 2310 of the 2312 schools that exist in the 34 counties of Santiago. With the schools and the block of the household georeferenced, we estimate the Euclidean distances between the centroid of the block and each of the 2310 schools.⁷ Ideally, distance could be measured by route as opposed to the Euclidian distance, but nonetheless, the distance presented "as the crow flies" serves its purpose and is easier to compute.⁸ In addition, the larger the block, the less accurate will be our measure.⁹ However, there is nothing suggesting that a systematic bias in this measure can exist, so using heteroskedastic robust methods should solve any eventual problem.

Fig. 1 shows the 34 municipalities of Santiago (Metropolitan Region) used and the georeferential information. The map on the left shows the blocks used from the CASEN survey. It can be seen that the survey covers the entire city with more than one survey carried out per block in most cases. The map on the right shows the concentration of schools per area. Each dot represents a school. It can also be seen that there is a relation between where schools are located and the concentration of blocks in each given area.

Table 1 shows some descriptive statistics of the students reported as attending school that lived in Santiago, according to the CASEN 2003 survey. The ratio of male to female students is similar in all cases, with each sex making up approximately 50% of students. Parents of students whose children attend private schools tend to have approximately 50% more years of schooling (on average) than those whose children attend municipal (public) schools, with parents of children in subsidized schools having just under the aver-

age number of schooling years in the case of fathers and just a little over the average in the case of mothers. There is an even greater difference in terms of income per capita. Parents whose children attend private schools often earn close to 3 times the average and close to five times the monthly wage of parents whose children attend municipal schools. Parents whose children attend subsidized schools earn below the average, but nonetheless earn about 50% more than those whose children attend municipal schools. Of all the students in the sample, 53% attended subsidized schools, almost 34% public schools, and 13% private schools.

Table 2 provides descriptive statistics (averages) of the distance, quality, and availability of different types of schools. Only 17.6% of the students attend to the nearest school. The percentage of students that attend public schools that also go to the nearest school is 24.4%. These shares decrease to 15.5% and 8.9% for students of subsidized and private schools respectively. If we consider schools of the same type, 36.2% of the students of public schools choose the nearest public school, and 24.3% (13.8%) of the students that attend subsidized (private) schools go the nearest school of the same type. Thus, choosing the school nearest to the household is not as prevalent as casual observers and survey suggest.

Children who attend municipal schools travel shorter distances than those who attend subsidized ones, with the average distances from their chosen schools being 2.57 and 2.78 km respectively, while children who attend private schools travel greater distances, with their chosen school at a distance of 4.22 km. Irrespective of the type of school, the average distance between the household and the nearest school is of approximately 0.5 km. Regarding nearby schools within a radius of 2 km, users of schools with vouchers make up a larger number of municipal and subsidized schools – approximately 20 – while users of private schools have an average of 8 schools with vouchers and 10 private schools within the same radius. Private schools tend to perform better than subsidized schools, and these do better than public schools in standardized tests. As expected, declared prices by parents, are higher for private and subsidized schools than municipal schools.

The average distance between the household and the school attended varies with the age of the child (Fig. 2). The schools chosen are nearer to the household in primary education and an almost discrete jump occurs when the student goes to high school (around age 15). This feature is related to the fact that there are fewer high schools than primary education schools, and that as age increases, the cost of traveling diminishes with age as students become more autonomous.

The same pattern is observed when considering the frequency of cases in which students choose the nearest school. Broadly speaking, irrespective of the type of school attended, there is a negative relationship between choosing the nearest school and the age of the student, particularly when the student reaches high school in the case of public and subsidized schools (Fig. 3).

⁷ Georeference means to define something, as a home or a school, in a physical space. In our case, it is establishing home and school locations in coordinates. This procedure is imperative to data modeling in the field of geographic information systems (GIS) and other cartographic methods. For example, household H will partner two coordinates (X_h , Y_h) representing its location on the map, in the same way, a school S has two coordinates (X_s , Y_s). Euclidean distance between home and school will be: $\sqrt{(X_h - X_s)^2 + (Y_h - Y_s)^2}$.

⁸ In addition, it is not possible to estimate real distance using real routes, because there are many zones without georeferenciation.

⁹ While most blocks have one hectare or less, and 82% have less than 3 ha, 8% are larger than 6 ha.

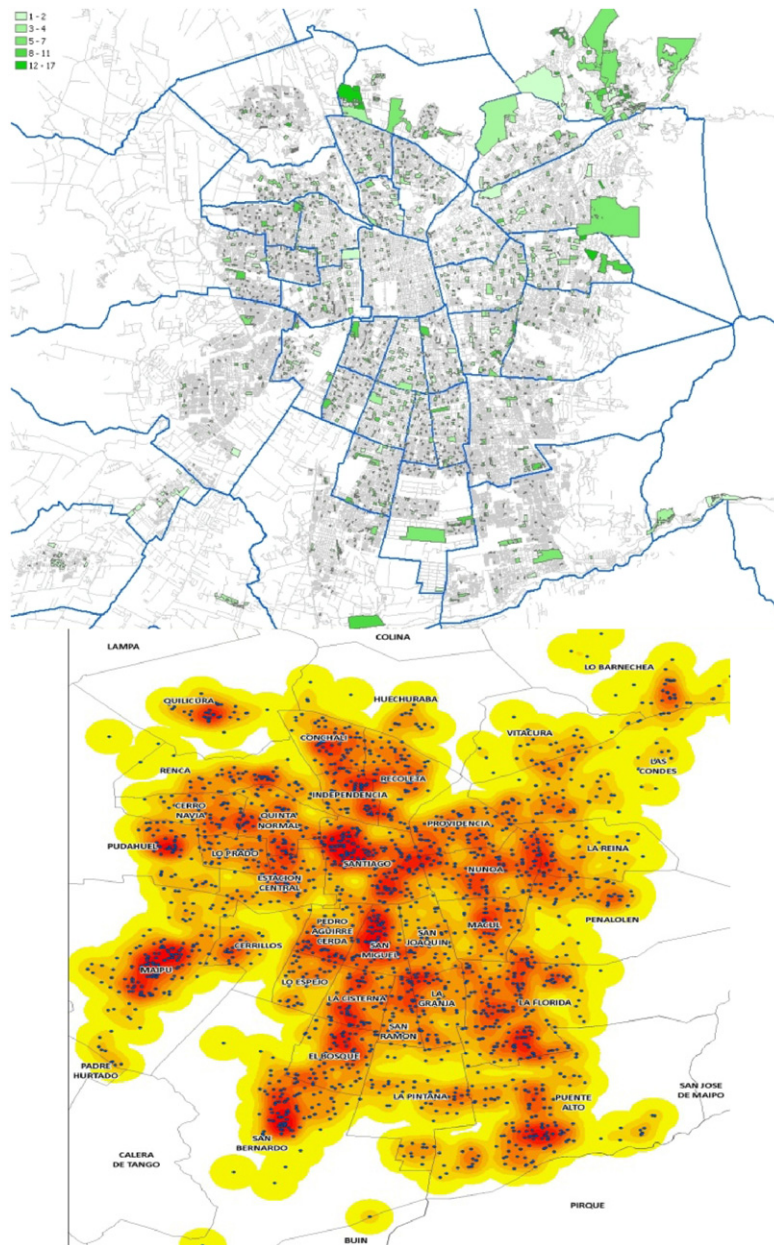


Fig. 1. Georeferencing of blocks and schools. Notes: (Left panel) Georeferencing of blocks. Green color denotes blocks where more surveys were conducted. (Right panel) Georeferencing of schools. Red color denotes greater concentration of schools. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

4. The empirical model

This section uses this information to evaluate the determinants of the choice of the nearest school, including distance. As said, parents consider several factors when choosing a school for their children. Some of them are specific to each child but common to every possible school selected (such as the age of the child, the education of the parents, gender of the child, income of the household, or other characteristics of the child or the household). Other

ers correspond to characteristics that are specific to each school and are common to every child and household (such as the type of school, its quality, its costs, and other characteristics of each school). Finally there are other attributes of each choice that are specific to the child and the school (most notably, the distance between the household and the school).

Let $i = 1, \dots, I$ index the individuals (students) in the sample and let $j = 1, \dots, J$ index the possible choices (schools). Denote by x_i to the vector of characteristics of

Table 2
Distance and quality by type of school administration (averages).

Variable	Total	Municipal	Subsidized	Private
Distance of school chosen	2.90	2.57	2.78	4.22
Quality of school chosen	256	240	257	296
Price of school chosen	31.68	5.59	22.19	135.34
Distance of nearest school	0.52	0.46	0.55	0.53
Quality of nearest school	248	241	247	272
Price of nearest school	29.79	17.28	22.03	92.26
Distance of nearest municipal school	0.90	0.67	0.93	1.34
Quality of nearest municipal school	232	230	231	246
Price of nearest municipal school	5.26	5.02	4.88	7.39
Distance of nearest subsidized school	0.78	0.71	0.73	1.15
Quality of nearest subsidized school	254	250	252	266
Price of nearest subsidized school	22.00	19.79	21.08	31.22
Distance of nearest private school	1.92	2.08	2.07	0.95
Quality of nearest private school	286	286	287	287
Price of nearest private school	98.18	96.41	91.23	130.30
Number of schools (2 km radius)	20.8	21.1	21.3	18.2
Quality of schools (2 km radius)	255	252	253	270
Number of municipal schools (2 km radius)	4.4	5.2	4.4	2.1
Quality of municipal schools (2 km radius)	241	239	240	254
Number of subsidized schools (2 km radius)	13.6	14.0	15.3	6.3
Quality of subsidized schools (2 km radius)	252	250	252	263
Number of private schools (2 km radius)	2.8	2.0	1.6	9.8
Quality of private schools (2 km radius)	286	285	286	287
Share of students that attend to				
Nearest school	17.6	24.4	15.5	8.9
Nearest school of the same type	26.9	36.2	24.3	13.8

Notes: Distance is measured in kilometers. Quality is measured as the average score of the students of the school on the standardized test. Price is expressed in US\$ per month. Weighted sample: 1,203,799.

the student and its household that do not depend on the school, by y_j to the vector of characteristics of the school that do not depend on the student, and by z_{ij} to the vector of attributes of the school that are specific to each student.

Define u_{ij} as the (indirect) utility of child i attending school j , so that:

$$u_{ij} = u(x_i, y_j, z_{ij}) + \varepsilon_{ij} \quad (1)$$

where $u(\cdot)$ corresponds to a systematic component and ε_{ij} is a (random) non-systematic component.

Given (1), agent i chooses school h if $u_{i,h} \geq u_{ij} \forall j \neq h$. Given a functional form for $u(\cdot)$ and a distributional

assumption of ε_{ij} parameters can be estimated using quasi-maximum likelihood (QML). When each individual has some factors that are specific to each choice (such as distance), the empirical literature tends to favor the use of conditional logit models for estimation.¹⁰

We take a different approach to evaluate the relevance of distance on school choice. Let d_{ij} denote the distance between household i and school j . Let d_{ni} be the distance between household i and the nearest school and u_{ni} the value of the utility function in (1) associated with choosing that school. On the other hand, let u_{mi} be the value of the utility function associated to the choice of the school that maximizes (1). Note that the school that minimizes d_{ij} and the one that maximizes u_{ij} may be different for each student i . Clearly, when the nearest school maximizes (1), u_{mi} and u_{ni} will coincide.

Let

$$v_i = \begin{cases} 1 & \text{if } u_{mi} = u_{ni} \\ 0 & \text{if } u_{mi} > u_{ni} \end{cases} \quad (2)$$

that is, v_i is the (observed) variable that takes the value of 1 when the student attends the school nearest to the household and 0 otherwise.

Evaluating (2) instead of (1) is convenient as now we can focus on modeling the determinants of choosing the

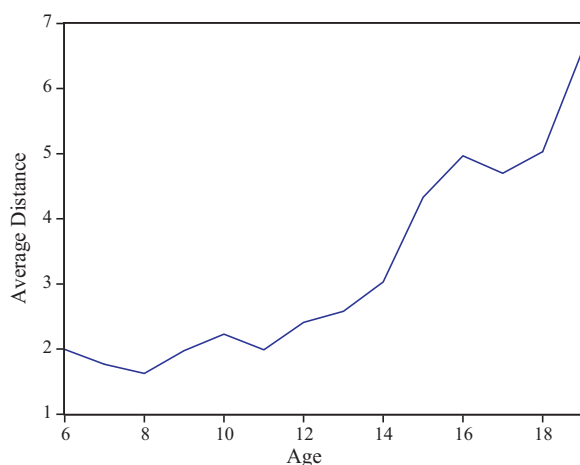


Fig. 2. Average distance (in km) by age.

¹⁰ In our case, the database consists of 2310 identified schools and 7479 students in the sample. Thus, in principle, more than 17 million distances would be needed to estimate this model.

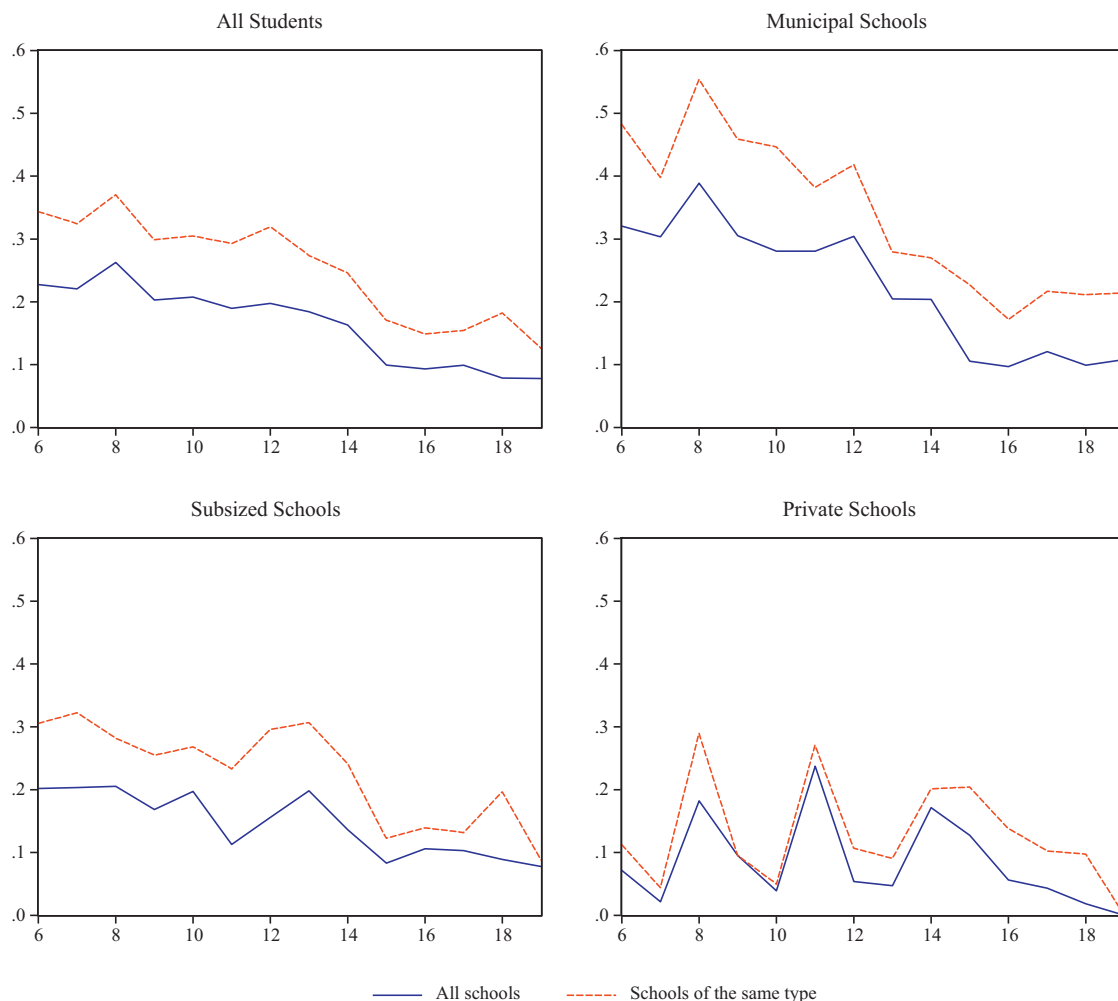


Fig. 3. Probability of choosing the nearest school by age.

nearest school using binary response models. The model considered is:

$$Pr(v_i = 1|w_i) = F(\beta'w_i)$$

where F is a postulated distribution function (say the standard normal), w_i is a vector of determinants, and β a vector of parameters to be estimated.¹¹

The vector of potential determinants considers, in the first place, individual or household characteristics obtained from the CASEN survey. Specifically, we consider gender of the student (1 = female), age of the student (in years), years of schooling of the father and/or mother, and Log of per capita income (logarithm of the ratio between total income of the household in US\$ and the number of mem-

bers of the household). In second place, we consider the characteristics of the nearest school and the school chosen. The nearest school is defined as the one that minimizes the Euclidean distance between a household and each of the 2310 schools identified. We consider the price charged by each school (measured in US\$), prices obtained from a survey conducted to parents of students taking the SIMCE test; the quality of each schools measured by its average score in the standardized tests (SIMCE), and the distance of these schools from the household measured in kilometers. In the third place, we consider the type of school chosen (public, subsidized, or private),¹² a competition variable defined by the number of schools in a 2 km radius from the centroid of the household, and a selection variable. As noted in Section 2, the empirical literature has stressed the importance of considering that schools may be the ones selecting the students and not parents selecting schools. To control for

¹¹ As considered in Ferreyra (2007), the choice of school and of residence may be jointly determined. While we have no data to control for the choice of residence, this may not be a prevalent problem in Chile because the vast majority of beneficiaries of the voucher program are from middle income and low income households. They tend to use publicly financed housing programs in which the location of the household is "exogenous" to them.

¹² It is not necessary to include distance of the school from the household, as this attribute defines the variable v in (2).

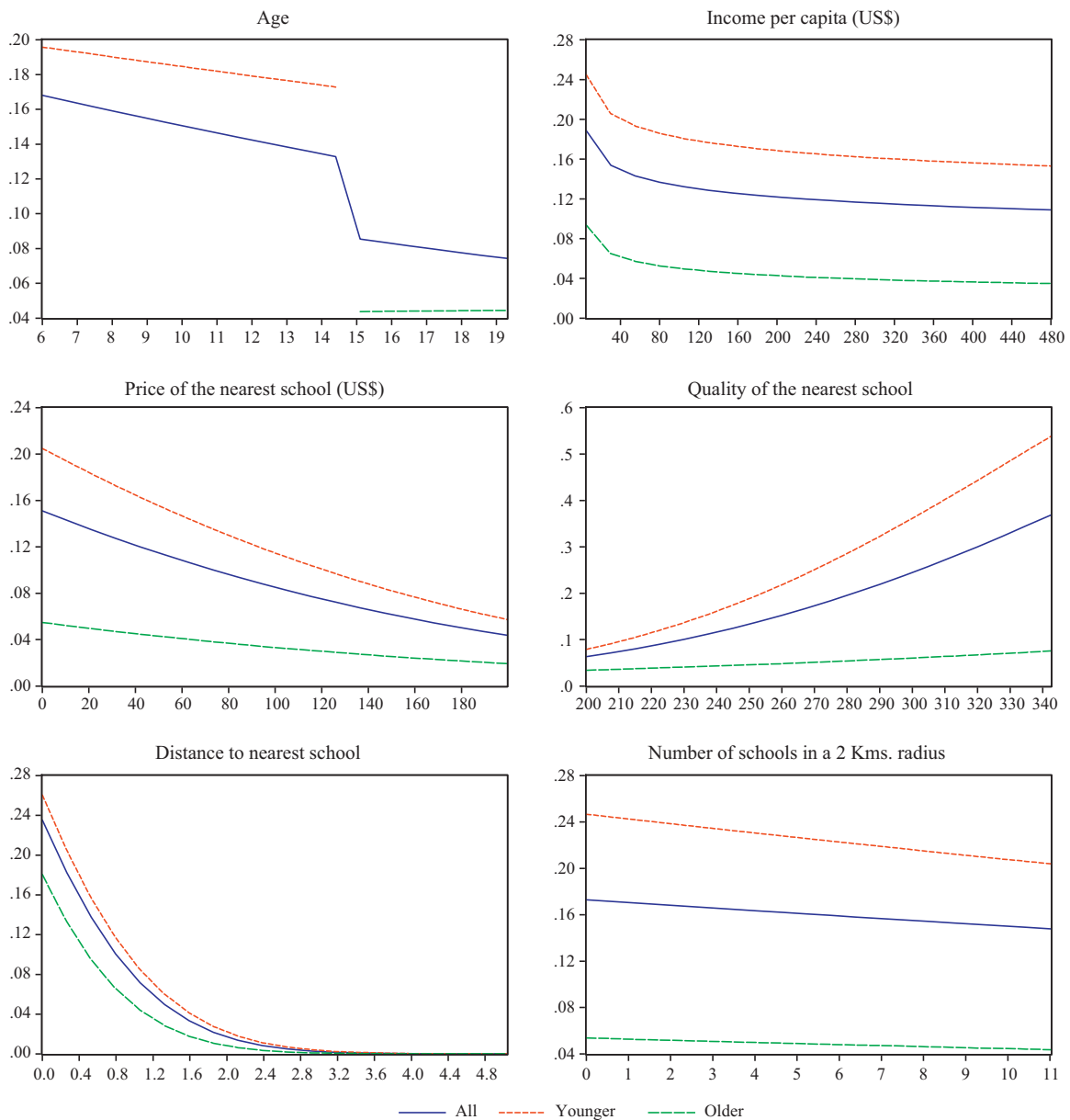


Fig. 4. Probability of choosing the nearest school by type of school.

this factor we use the inverse Mills ratios for the school that the child attends estimated by García and Paredes (2010). A detailed explanation of the construction of this variable follows.

Schools use different selection methods that depend not only on characteristics specific of the school but also of the student. For instance, some schools conduct admission tests intended to measure student's knowledge and abilities which are positively correlated with the family's background. A natural consequence of such selection tests is that those schools that require them have non-representative populations.

To control for the possible selection by schools, we follow Heckman's (1979) two-step method. In the first

place, we distinguish four selection criteria according to the requirements imposed by the school: religious reasons, social economic level, cognitive ability and family values. This information is obtainable from questionnaires in which parents indicate if a particular school requires that the student be baptized, a certification of solvency (parents' income), if the student must take an entrance exam, or if there is a interview to the parents.

We define that a school selects students under different criteria, as for instance, when at least one or two forms of selection were present. We then compute the probability P_{ijk} that a student i is selected by school j , using the method k . From the imputed probability, we compute the inverse of the Mill's ratio, the control variable, which is

defined as:

$$\lambda_{ijk} = \frac{\phi(\Phi^{-1}(P_{ijk}))}{P_{ijk}}$$

where ϕ is standard normal density, Φ the cumulative normal distribution, and P_{ijk} is the estimated probability that student i would be accepted in school j which selects using the selection process k .¹³

Going back to the general discussion of the model, the attribute (distance) of each possible choice (school) is specific to each individual (student). The conventional framework in these cases would require modeling the probabilities of choosing each school for each student. This path would not only be computationally demanding, but would make the results difficult to interpret. Our framework directly tackles the question at hand (what are the determinants of choosing the nearest school) and also allows us to use conventional models for binary choice models. Thus, the model assumes households compare the characteristics of the school actually chosen (which may or not coincide with the nearest school available), and the nearest school. Our choice reflects that when the household chooses a different school from the nearest school, we need to compare its characteristics with the one actually chosen, because that is (presumably) the relevant comparison by the household.

Thus, our interpretation of the results is natural. It is expected that the better the attributes of the nearest school (in terms of quality, distance or charge), the more likely it should be that it would be chosen. The probability of choosing the nearest school should also depend on characteristics of the student. When traveling is more costly, for example in terms of safety or autonomy, parents should be more likely to choose the nearest school. On the other hand, it should be less likely to choose the nearest school with better characteristics of alternative schools (in terms of price or quality). In addition, a higher level of education on the parents' part and a greater income in the household should increase the set from which to choose schools and make it less likely to choose the nearest school. Finally, when there are more alternatives available (more schools nearby), choosing the nearest school should be less likely.

Table 3 presents the results of estimating a probit model for this choice.¹⁴ The results show that households would (marginally) prefer a closer school if the child is female. Consistent with the evidence of Fig. 2, the model also shows that, the older the child, the lower the probability of choosing the nearest school and that there is a discrete decrease in the probability of choosing the nearest school when the student attends high school (when the child reaches the age of 15). The model also shows that the probability of choosing the nearest school decreases with household income and the mother's education.¹⁵ Thus, as the education of the

parents or income increases, the less likely the family is to choose the nearest school.

Increasing the number of schools near the household reduces the probability of attending the nearest school. As would be expected, households are more likely to choose the nearest school when its quality is higher, when its price is lower, or when it is closer.¹⁶ Thus, as economic theory predicts, the model shows that there is indeed a trade-off between quality, costs, and distance that parents consider. Consistently, the better the quality of the school chosen, the less likely it is that the student would attend the nearest school. Also, as shown in Fig. 3, students attending public schools are more likely to attend the nearest school.

These results are consistent with Hastings et al. (2006), that conclude that parents value proximity and schools' average test scores. However, they differ from Gallego and Hernando (2008), since we do find that family income and the educational level of the parents do have an effect on school choice.

The model reported in Table 3 is estimated considering all the students. It may be argued that parents who choose (say) a public school are less likely to consider a private school as a close substitute. While such separation may be important for distinguishing private and voucher (public and private) schools, it is not that obvious when comparing public and private voucher schools. Consequently, we also estimated the model reported in Table 3 for the case of municipal, subsidized and private schools. In addition, we distinguished students by gender, and age. Regardless of the type of school, the results are broadly consistent with what was presented in Table 3.¹⁷

Table 4 and the panels of Fig. 4 show how changes in different factors affect the probability of choosing the nearest school by the age of the student.¹⁸ Apart from the scale (children who attend municipal schools are more likely to attend the nearest school), the probability of attending the nearest school is also more responsive to changes in all factors for younger students. Particularly important in this case is having more options available near the house.

One by-product of the model estimated, is that rates of substitution between two variables can be readily obtained. For example, consider the case in which the quality of the nearest school increases by standard deviation. According to our model, this would increase the probability that a family chooses it. As the probability of attending the nearest school is decreasing in its price, we can compute how much the price of the nearest school should decrease in order to match the increased probability of attending the nearest school due to the increased quality. In this way, we

¹³ The results reported below are robust to using different lambdas or the exclusion of selection as a control variable.

¹⁴ Marginal effects are computed as the average of marginal effects for each observation.

¹⁵ Models that also include the education of the father have similar results and are available upon request. However, as there are many

missing values on this variable, we report the results of using only the education of the mother.

¹⁶ Distance to the school chosen cannot be included in the model as distance is precisely the attribute used to construct (2). When a household actually chooses the nearest school, all attributes (excluding distance) would coincide. In such case, the relevant factors for determining the choice would be most student and household specific attributes.

¹⁷ Parameter estimates for type of school and gender are available upon request.

¹⁸ The probabilities were evaluated in the means of the independent variables corresponding to each sample. Then the variable to be considered was modified.

Table 3

Probit model for choosing the nearest school.

Variable	Estimate	Marginal effect
Constant	–0.2523 (0.0264)	
Gender (1 = woman)	0.0417 (0.0029)	0.0097 (0.0013)
Age	–0.0180 (0.0006)	–0.0042 (0.0011)
Older than 14	–0.2439 (0.0056)	–0.0566 (0.0018)
Schooling of mother	–0.0131 (0.0005)	–0.0030 (0.0011)
Log of income per capita	–0.0766 (0.0021)	–0.0178 (0.0012)
Price (school chosen)	0.0035 (0.0001)	0.0008 (0.0011)
Price (nearest school)	–0.0034 (0.0001)	–0.0008 (0.0011)
Quality (school chosen)	–0.0062 (0.0001)	–0.0014 (0.0011)
Quality (nearest school)	0.0084 (0.0001)	0.0019 (0.0011)
Distance (nearest school)	–0.6998 (0.0062)	–0.1625 (0.0019)
Number of schools	–0.0094 (0.0002)	–0.0022 (0.0011)
Type (1 = public school)	0.0536 (0.0038)	0.0125 (0.0014)
Lambda (selection)	–0.2337 (0.0030)	–0.0543 (0.0013)
Observations = 1,114,061	LRT = 92,936 [0.00]	McFadden's pseudo R^2 = 0.090

Notes: Standard deviations in parenthesis. LRT = Likelihood Ratio Test for the null hypothesis that all coefficients (except for the constant) are 0. *P*-value in brackets.

Table 4

Probit model for choosing the nearest school.

Variable	Younger	Marginal	Older	Marginal
Constant	–0.8953 (0.0400)		–0.2807 (0.0641)	
Gender (1 = woman)	0.0142 (0.0042)	0.0036 (0.0015)	0.1374 (0.0066)	0.0218 (0.0024)
Age	–0.0180 (0.0008)	–0.0046 (0.0012)	0.0034 (0.0025)	0.0005 (0.0022)
Schooling of mother	–0.0071 (0.0005)	–0.0018 (0.0012)	–0.0334 (0.0010)	–0.0053 (0.0022)
Log of Income per capita	–0.0730 (0.0024)	–0.0188 (0.0014)	–0.1089 (0.0044)	–0.0173 (0.0023)
Price (school chosen)	0.0031 (0.0001)	0.0008 (0.0011)	0.0065 (0.0002)	0.0010 (0.0022)
Price (nearest school)	–0.0038 (0.0001)	–0.0010 (0.0011)	–0.0024 (0.0001)	–0.0004 (0.0022)
Quality (school chosen)	–0.0084 (0.0001)	–0.0022 (0.0011)	–0.0020 (0.0002)	–0.0003 (0.0022)
Quality (nearest school)	0.0106 (0.0001)	0.0027 (0.0011)	0.0027 (0.0002)	0.0004 (0.0022)
Distance (nearest school)	–0.6885 (0.0076)	–0.1776 (0.0024)	–0.7492 (0.0111)	–0.1191 (0.0028)
Number of schools	–0.0100 (0.0002)	–0.0026 (0.0011)	–0.0070 (0.0005)	–0.0011 (0.0022)
Type (1 = public school)	0.0280 (0.0044)	0.0072 (0.0017)	0.1124 (0.0077)	0.01789 (0.0025)
Lambda (selection)	–0.2224 (0.0033)	–0.0574 (0.0015)	–0.3277 (0.0093)	–0.0521 (0.0027)
Observations	807,126		306,935	
Pseudo- R^2	0.079		0.072	

Notes: Standard deviations in parenthesis. LRT = Likelihood Ratio Test for the null hypothesis that all coefficients (except for the constant) are 0.

can compute the amount of money that a family would be willing to sacrifice in order to obtain greater quality. The same reasoning can be applied to evaluate the substitution between price of the nearest school and its distance.

Based on the results of Table 3, students would be willing to “walk” 0.2 standard deviations of distance (0.236 km) to avoid paying one standard deviation more of the price of the nearest school (US\$ 48). Furthermore, parents would be willing to pay 1.5 standard deviation of the price (US\$ 74) more to attend the nearest school if it improved in one standard deviation its test (30 points).¹⁹

Our results are consistent with surveys that indicate that parents value distance as an important characteristic of the school, but also consistent with families valuing quality.

¹⁹ The gap between the average score of private and public schools is between 40 and 50 points (in favor of the former). Chumacero and Paredes (2008) find that (after controlling for other characteristics) the gap between private and municipal schools can be of up to 15 points (half a standard deviation). Thus, according to this estimate, roughly 40% of the gap between the average price of public and private schools can be accounted for with this valuation.

5. Concluding remarks

Although Chile is the country where the educational voucher system is most widely used, and its application is more in line with Friedman's original idea, many of the recommendations given to improve it come from anecdotal evidence including surveys which suggest that people do not consider the quality of education at the moment of choosing a school and consequently that the system does not provide incentives for competition.

We use a new database that accurately estimates the distance between the household and school to test how important location, price, quality, and competition are when choosing schools. We also developed a novel specification to address the issue of how parents choose the school their children attend. This specification allows us to use conventional discrete choice models that, in their absence, would require computationally intensive and cumbersome methods.

The model estimated considers characteristics of the students, the households, and the schools. We show that location, price, quality, and potential competition are all relevant determinants of the choice of schools and quan-

tify the implied trade-offs. Our results indicate that both quality and distance are highly valued by households.

While we do not evaluate the impact of the voucher system on academic performance, this paper is useful because it provides robust evidence that contradicts claims that would undermine the voucher system as a means to improving quality through competition. It is also useful as it presents estimates of how valuable quality (particularly for low income families) is. It also suggests that it would probably be more desirable to increase the portability of the voucher (as well as its amount) instead of focusing on targeting (centrally) improving the quality of public schools in a given area.

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